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Normalizing Flow-Based Modeling for Detecting Ionizing Sources within H II Regions

Master's Thesis

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*“The most incomprehensible thing about the Universe
is that it is comprehensible.”*

– Albert Einstein

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Abstract

The Epoch of Reionization marks a pivotal stage in cosmic history, as the first luminous sources (i.e. primordial stars and galaxies) ionized the neutral hydrogen in the intergalactic medium. In this thesis, we develop a conditional Normalizing Flow framework to model the distribution of ionizing sources within H II regions (“bubbles”). We demonstrate that our method outperforms simpler state-of-the-art approaches based on source placement at the innermost points. Quantitative evaluations show a consistently high R^2 -value at $\gtrsim 80\%$ across different shapes, volumes, and reionization scenarios. Furthermore, our model can infer the total halo mass inside a bubble purely from its geometry. Tests on noisy data also show promising results under simulated SKA-Low observation noise level.

By using the incoming data from SKA-Low, the model can be used as a pointer to JWST observation. This work aims to establish a connection between the nature of primordial luminous sources and the Cosmic Epoch of Reionization.

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The 21-cm Line and Cosmic Reionization

The Epoch of Reionization (EoR) represents a pivotal period in the history of the Universe, during which the first luminous sources (i.e. primordial stars and galaxies) ionized the neutral hydrogen that filled the intergalactic medium (IGM). Observationally, one of the most promising probes of this era is the 21-cm transition line of hydrogen. By monitoring fluctuations in the brightness temperature of the 21-cm signal across different redshifts, one can reconstruct, how neutral hydrogen was distributed in the early Universe and, by implication, how reionization proceeded (e.g., [Barkana & Loeb 2001](#); [Furlanetto et al. 2006](#); [Pritchard & Loeb 2012](#); [Zaroubi 2013](#); [Ferrara & Pandolfi 2014](#)). In the context of this thesis, the focus is on understanding and modeling the density distribution within ionized H II regions (in this thesis often called “bubbles”) in the early Universe. These regions are formed around luminous sources of radiation, gradually affecting the IGM and transforming it from predominantly neutral to predominantly ionized. This chapter provides the theoretical background on the physics of the 21-cm line during EoR (Section 1.1) and the properties of ionized H II regions during this epoch (Section 1.2).

1.1 The 21-cm Line in Cosmology

1.1.1 The Hyperfine Transition

Neutral hydrogen (HI) possesses a ground state split by a hyperfine interaction between the spin of the proton and the spin of the electron (see Figure 1.1). The energy difference between the triplet and the singlet state corresponds to a wavelength of $\lambda \approx 21$ cm and frequency $\nu \approx 1420$ MHz (e.g., [Ewen & Purcell 1951](#)). Though this transition is highly forbidden (with a spontaneous emission timescale of around 10^7 years), the sheer abundance of hydrogen in the Universe ensures that the 21-cm line is observable on cosmological scales.

The spin temperature T_s determines the abundances of the two hyperfine levels and is defined through the Boltzmann equation ([Liu et al. 2022](#)):

$$\frac{n_1}{n_0} = 3 \exp\left(-\frac{h\nu_{10}}{k_b T_s}\right), \quad (1.1)$$

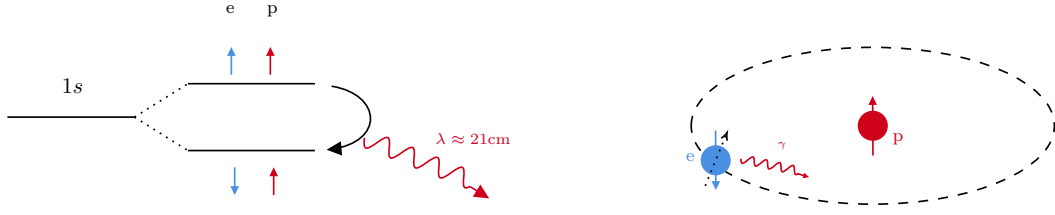


Figure 1.1: The hydrogen atom in its ground state (1s) experiences hyperfine splitting due to the proton (p^+) and electron (e^-) spins. When the spins are aligned (triplet state), the atom is in a slightly higher energy state than the anti-aligned configuration (singlet state). The transition from the triplet to the singlet state occurs via the spontaneous emission of a photon with a wavelength of approximately $\lambda \approx 21$ cm and frequency $\nu \approx 1420$ MHz.

where n_1 and n_0 are the number densities in the triplet and singlet states, h is Planck's constant, k_b is Boltzmann's constant, and $\nu_{10} \approx 1420.406$ MHz is the frequency of the 21-cm-line photon, which is emitted during the spin-flip. The factor 3 accounts for the statistical degeneracy of the triplet state. The energy difference between the two hyperfine levels is given by $k_B T_* = h\nu_{10} \approx 5.8598 \times 10^{-6}$ eV (e.g., [Furlanetto et al. 2006](#); [Liu et al. 2022](#)).

1.1.2 Brightness Temperature

The observable 21-cm signal is characterized by the differential brightness temperature δT_b relative to the cosmic microwave background temperature ([Mellema et al. 2013](#); [Pritchard & Loeb 2012](#); [Zaroubi 2013](#)):

$$\delta T_b(\nu) \approx 27 x_{\text{HI}} (1 + \delta_b) \sqrt{\frac{1+z}{10}} \left(1 - \frac{T_\gamma(z)}{T_s} \right) \left(\frac{\Omega_b}{0.044} \frac{h}{0.7} \right) \left(\frac{\Omega_m}{0.27} \right)^{-1/2} \times \left(\frac{H(z)/(1+z)}{dv_{\parallel}/dr_{\parallel}} \right) \text{ mK}, \quad (1.2)$$

where x_{HI} is the neutral hydrogen fraction, $(1+\delta_b)$ is the fractional overdensity of baryons, z is the redshift, T_γ is the CMB temperature, $H(z)$ is the Hubble parameter, and $dv_{\parallel}/dr_{\parallel}$ is the gradient of the line-of-sight velocity with respect to the line-of-sight comoving distance. It accounts for two effects: the Hubble expansion $H(z)/(1+z)$ and peculiar velocities $v_{p,\parallel}$:

$$\frac{dv_{\parallel}}{dr_{\parallel}} = \frac{H(z)}{1+z} + \frac{dv_{p,\parallel}}{dr_{p,\parallel}} \quad (1.3)$$

Without peculiar velocities, we can set the last term in Equation 1.2 equal to 1. During reionization, x_{HI} transitions from nearly 1 (entirely neutral) to nearly 0 (fully ionized), and thus, δT_b (which, according to Equation 1.2, obeys $\delta T_b \propto x_{\text{HI}}$) serves as a

powerful probe of the neutral hydrogen distribution in the early Universe. It is around a few tens to hundreds of mK before reionization and $\delta T_b = 0$ after reionization.

1.1.3 Ongoing Radio and Infrared Experiments

The Square Kilometre Array (SKA) is an ongoing radio interferometry telescope. Its low-frequency component, SKA-Low, is located in Western Australia. It will observe patches of the sky at different frequencies, ranging from 50 MHz ($z \sim 27$) to 350 MHz ($z \sim 3$).

SKA-Low will be sensitive enough not only to detect the global signal of the differential brightness (Equation 1.2) but also to create images of the fluctuations in the redshifted 21-cm signal at different observed frequencies/redshifts (Mellema et al. 2013; Wyithe et al. 2015; Koopmans et al. 2015). Moreover, it will observe a large portion of the sky with an estimated Field of View (FoV) of 100 deg^2 and angular resolution of 14 arcsec .

In recent years, the James Webb Space Telescope (JWST) has provided direct observations of bright UV-luminous sources at high redshifts, $z \geq 8$ (Naidu et al. 2022; Bouwens et al. 2023a b; Harikane et al. 2023; Leung et al. 2023; Pérez-González et al. 2023; McLeod et al. 2023; Robertson et al. 2024; Willott et al. 2024), offering a unique insight into early Universe star and galaxy formation connected to reionization. Eventually, one would like to map the entire sky at different redshifts. However, due to the nature of the experiment, the near-infrared camera (NIRCam) on JWST has a field of view of only 132 arcsec per side (Cox & Lallo 2017; Sahlmann et al. 2019). Therefore, a complete sky map would require several hours of observation, considerable data and effort.

Consequently, we can expect these two experiments to work in symbiosis. We expect ionizing sources to be located inside ionized regions (bubbles), where SKA-Low together with our model serve as a pointer to JWST observations. We hope that this could help us connect the nature of the primordial luminous sources and the Cosmic EoR.

1.2 Formation and Evolution of H II Bubbles

This section closely follows the approach of Park et al. 2020 and Giri et al. 2024. As the first ionizing sources form, they create expanding H II regions. Several key factors determine the properties of these ionized bubbles: the ionizing source properties, the density distribution in general, and also the clumpiness of the gas, as substructures within the IGM can alter the ionization fronts. At early times ($z \approx 20$), only small, isolated ionized regions existed, as the number of ionizing sources was low. As more galaxies formed and ionizing radiation increased, the bubbles expanded and started to overlap ($z \approx 15$). By $z \approx 10$, ionized regions merged, eventually leading to the complete reionization of the IGM at $z \approx 6$. This process is shown schematically in Figure 1.2.

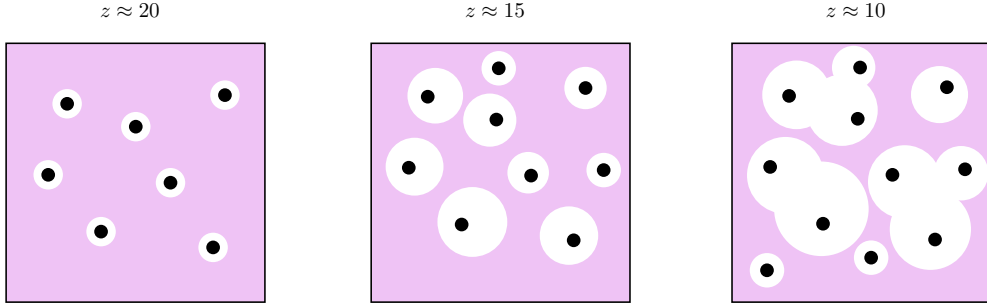


Figure 1.2: Schematic illustration of the cosmic reionization process at different redshifts. The purple background represents neutral hydrogen (H I), while the white regions correspond to ionized bubbles (H II) surrounding ionizing sources (black dots). At $z \approx 20$, only small ionized regions have formed around early galaxies. By $z \approx 15$, these ionized bubbles have grown, though large portions of the universe remain neutral. At $z \approx 10$, the bubbles have expanded and begun to merge, indicating the progression toward full reionization.

1.2.1 Ionizing Photon Production Rate

The expansion of an H II region is directly related to the rate at which ionizing photons are produced and escape into the surrounding IGM. The number of ionizing photons per second escaping into the IGM is given by

$$\dot{N}_\gamma(M_h) = f_{\text{esc}}(M_h) \left(\frac{N_{\text{ion}}}{m_p} \right) \dot{M}_\star(M_h), \quad (1.4)$$

where $f_{\text{esc}}(M_h)$ is the escape fraction of ionizing photons, N_{ion} is the number of ionizing photons produced per unit proton mass in stars, m_p is the proton mass, and $\dot{M}_\star(M_h)$ is the rate at which stellar mass is formed. Since star formation occurs within galaxies, and these galaxies reside in dark matter halos, the stellar mass growth rate $\dot{M}_\star(M_h)$ is expected to follow the halo accretion rate:

$$\dot{M}_\star(M_h) = f_\star(M_h) \frac{\Omega_b}{\Omega_m} \dot{M}_h, \quad (1.5)$$

where $\frac{\Omega_b}{\Omega_m}$ is the fraction of baryonic mass in matter, $f_\star(M_h)$ is the fraction of the halo's baryonic mass actually converted into stars, and $\dot{M}_h$ is the halo mass accretion rate. The fraction of baryons forming stars is typically assumed to follow a power law:

$$f_\star(M_h) = f_{\star,0} \left(\frac{M_h}{10^{10} M_\odot} \right)^{\alpha_\star}, \quad (1.6)$$

1 The 21-cm Line and Cosmic Reionization

where $f_{\star,0}$ and $\alpha_\star$ are constants. The ionizing photon escape fraction $f_{\text{esc}}(M_h)$ also depends on the halo mass and follows a similar power-law dependence:

$$f_{\text{esc}}(M_h) = f_{\text{esc},0} \left(\frac{M_h}{10^{10} M_\odot} \right)^{\alpha_{\text{esc}}}. \quad (1.7)$$

These relations determine how efficiently ionizing photons reach the IGM and contribute to the growth of H II regions.

The total number of ionizing photons produced over time depends on the growth of dark matter halos. The halo accretion rate, $\dot{M}_h$, follows an approximate growth model, similar to Equation (3) in [Park et al. \(2019\)](#):

$$\dot{M}_h(z) = \frac{M_h(z)}{[\alpha_h (1+z) H(z)]^{-1}}, \quad (1.8)$$

where $H(z)$ is the Hubble parameter, and α_h is a parameter that regulates the characteristic timescale of halo growth. From this relation, the ionizing photon production rate can be written as:

$$\dot{N}_\gamma(M_h, z) = \frac{N_{\text{ion}} f_{\text{esc},0} f_{\star,0}}{m_p} \frac{\Omega_b}{\Omega_m} \alpha_h (1+z) H(z) M_h(z) \left(\frac{M_h(z)}{10^{10} M_\odot} \right)^{\alpha_{\text{esc}} \alpha_\star}. \quad (1.9)$$

For instance, adopting the **Source1_SinkA** (**S1SA**) modelⁱ from [Giri et al. \(2024\)](#) with $\alpha_h = 0.79$, $N_{\text{ion}} f_{\text{esc},0} f_{\star,0} = 4$, and $\alpha_{\text{esc}} \alpha_\star = 0$ yields a simpler relation,

$$\dot{N}_{\gamma, \text{S1SA}}(M_h, z) = f_\gamma(z) \frac{\Omega_b}{\Omega_m} M_h(z), \quad \text{with } f_\gamma(z) = \frac{3.16 (1+z) H(z)}{m_p}. \quad (1.10)$$

Hence, as halos grow in mass, their photon production rate naturally rises, driving the gradual expansion of ionized regions.

In summary, this chapter introduced the 21-cm line as a powerful probe of cosmic reionization, detailing how hydrogen's hyperfine transition and the resulting brightness temperature fluctuations contribute to the early universe's hydrogen distribution. We then examined the formation and evolution of ionized H II bubbles and the role of ionizing photon production $\dot{N}_\gamma$. The next chapter of this thesis will now explore a simulation-driven deep learning approach to modeling the structures inside H II regions using Normalizing Flows.

ⁱ The source numbers (**S1**, **S2**, **S3**) refer to different parameterizations of the ionizing photon escape fraction $f_\star$, while the sink models control the mean free paths of the ionizing photons in the IGM before being absorbed by photon sinks. **SinkA** (**SA**) simulates a universe with more dense absorbers, which slows reionization and keeps ionized bubbles smaller. **SinkB** (**SB**) allows for faster reionization and larger ionized regions due to a longer photon travel distance. **SinkC** (**SC**) is essentially a more extreme version of **SinkA** (**SA**) — it assumes an even denser or more clumpy IGM, further limiting how far photons can travel before being absorbed.

Normalizing Flows

Over the past few years, machine learning methods have gained traction in astrophysics, contributing to progress across multiple research areas (e.g., [Guzman & Meyers 2021](#); [Jeffrey et al. 2020](#); [Vafaei Sadr & Farsian 2021](#); [Yoshiura et al. 2021](#)). Among these methods, Normalizing Flows have emerged as a powerful class of generative models, that transform a simple (often referred to as “base”) probability distribution into a more complex one. They have gained popularity due to their ability to perform exact and tractable density estimation, while at the same time allowing efficient sampling (e.g., [Rezende & Mohamed 2015](#); [Dinh et al. 2015](#); [Kobyzev et al. 2021](#)). Applications of Normalizing Flows include image generation (e.g., [Ho et al. 2019](#); [Kingma & Dhariwal 2018](#)), noise modelling (e.g., [Abdelhamed et al. 2019](#)), video generation (e.g., [Kumar et al. 2019](#)), audio generation (e.g., [Esling et al. 2019](#); [Kim et al. 2018](#); [Prenger et al. 2019](#)), reinforcement learning (e.g., [Mazouze et al. 2019](#); [Ward et al. 2019](#); [Touati et al. 2019](#)), computer graphics (e.g., [Müller et al. 2018](#)), and different areas in physics (e.g., [Kanwar et al. 2020](#); [Köhler et al. 2019](#); [Noé et al. 2019](#); [Wirnsberger et al. 2020](#); [Wong et al. 2020](#)).

The key principle behind Normalizing Flows is to start from a simple base distribution (e.g. a standard Gaussian) and then apply a sequence of invertible transformations (i.e. flows) to morph it into a target distribution. In flow-based generative modeling, each flow is carefully designed so that:

1. It is *differentiable* almost everywhere.
2. It is *invertible*, allowing transformations in both directions: from data space to latent space and vice versa.
3. The *Jacobian* of the mapping (or its log-determinant) is *computationally tractable*, so one can directly optimize the model via maximum likelihood.

Unlike Variational Autoencoders (VAEs) or Generative Adversarial Networks (GANs), Normalizing Flows allow exact computation of a probability density at any point in the data space.

2.1 Mathematical Foundation

Let $\mathbf{Z} \in \mathbb{R}^D$ be a random variable with a known and tractable probability density function $p_{\mathbf{Z}}(\mathbf{z})$. Suppose we have a differentiable, invertible function

$$g : \mathbb{R}^D \rightarrow \mathbb{R}^D, \quad (2.1)$$

and its inverse

$$f = g^{-1} : \mathbb{R}^D \rightarrow \mathbb{R}^D. \quad (2.2)$$

Define the target random variable $\mathbf{Y}$ via $\mathbf{Y} = g(\mathbf{Z})$. Then by the change of variables formula, the distribution of $\mathbf{Y}$ satisfiesⁱ

$$p_{\mathbf{Y}}(\mathbf{y}) = p_{\mathbf{Z}}(f(\mathbf{y})) \left| \det \frac{\partial f(\mathbf{y})}{\partial \mathbf{y}} \right|, \quad (2.5)$$

where $\frac{\partial f(\mathbf{y})}{\partial \mathbf{y}}$ denotes the Jacobian matrix of f with respect to $\mathbf{y}$. When g is composed of multiple invertible layers $g_1, g_2, \dots, g_N$, for each mapping $\mathbb{R}^D \rightarrow \mathbb{R}^D$:

$$g(\mathbf{z}) = g_N \circ g_{N-1} \circ \dots \circ g_1(\mathbf{z}), \quad (2.6)$$

the result is still an invertible transformation. The log-likelihood of data $\mathbf{y}$ under the model can be written:

$$\log p_{\mathbf{Y}}(\mathbf{y}) = \log p_{\mathbf{Z}}(f(\mathbf{y})) + \sum_{k=1}^N \log \left| \det \frac{\partial f_k(\mathbf{x}_k)}{\partial \mathbf{x}_k} \right|, \quad (2.7)$$

where $f_k = g_k^{-1}$ and $\mathbf{x}_k = g_k \circ g_{k-1} \circ \dots \circ g_1(\mathbf{z})$ is the intermediate representation.

In practice, we do not directly train g , but rather its inverse $f = g^{-1}$. The goal is to adjust the parameters of f such that, when applied to data $\mathbf{Y}$, it maps them to the simple base distribution $p_{\mathbf{Z}}(\mathbf{z})$ (we call this *normalizing*; see Figure 2.1). Training involves learning the parameters of the transformations in f such that the total likelihood of the observed data is maximized.

Some basic types of Normalizing Flows include planar and radial flows (Rezende & Mohamed 2015). The main challenge in designing these flows is balancing expressive power with efficient computation of the Jacobian determinant. One solution, introduced in NICE (Dinh et al. 2015), is the use of affine coupling layers. These layers split the input, transforming only part of it while keeping the rest unchanged. Real NVP (Dinh et al.

ⁱThis follows from the fact that

$$d\mathbf{y} = \left| \det \frac{\partial g(\mathbf{z})}{\partial \mathbf{z}} \right| d\mathbf{z} \quad (2.3)$$

and

$$\int p_{\mathbf{Z}}(\mathbf{z}) d\mathbf{z} = 1 = \int p_{\mathbf{Y}}(\mathbf{y}) d\mathbf{y}. \quad (2.4)$$

2 Normalizing Flows

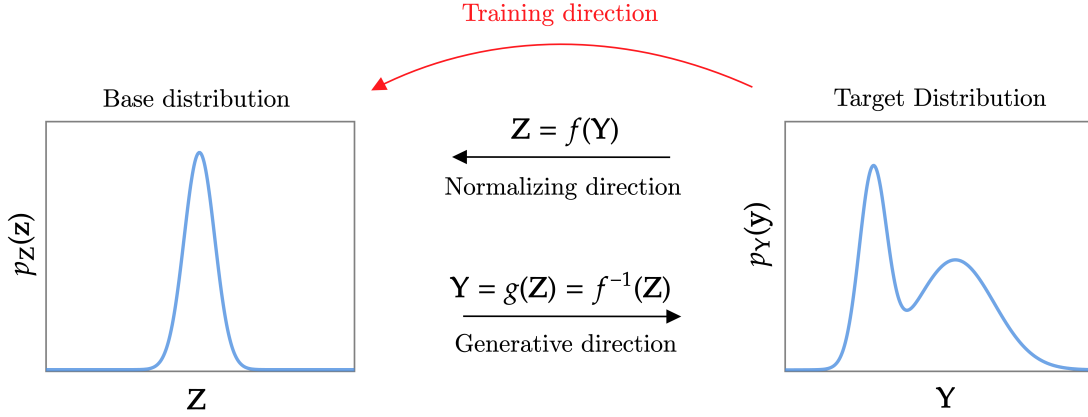


Figure 2.1: Change of Variables (Equation (2.5)). Left: Probability density function of the base distribution $p_{\mathbf{Z}}$. Right: Probability density function of the target distribution $p_{\mathbf{Y}}(\mathbf{y})$. There exists a bijective transformation g such that $\mathbf{Y} = g(\mathbf{Z})$, with inverse mapping $f = g^{-1}$, which is the function we train.

2017) improved on this by adding scale and translation parameters, making the flow more complex. A key advantage of Real NVP is that its Jacobian has a lower triangular structure, allowing for efficient computation of the log-determinant in just $\mathcal{O}(D)$ time.

2.2 GLOW Architecture

The GLOW (**G**enerative **F**low) architecture builds upon RealNVP by introducing an invertible 1×1 convolution. To extend RealNVP into GLOW (Kingma & Dhariwal 2018), we make each layer g_k a composition of three invertible transformations: an activation normalization (Act-Norm), an invertible 1×1 convolution, and an affine coupling. In symbolic terms, a single Glow step can be drawn as in Figure 2.2.

Act-Norm enforces that the outputs of this block have zero mean and unit variance (per dimension). Mathematically, for each component $i = 1, \dots, D$, we define

$$y_i = \exp(\alpha_i) x_i + \beta_i, \quad (2.8)$$

where α_i and β_i are learned parameters. The log-determinant of the Jacobian for this transformation is then simply given by

$$\log \left| \det \frac{\partial \mathbf{y}}{\partial \mathbf{x}} \right| = \sum_{i=1}^D \alpha_i. \quad (2.9)$$

The invertible 1×1 convolution replaces the simple channel-permutation used in RealNVP with a learned linear transformation that acts on the channel dimension. If we write x

2 Normalizing Flows

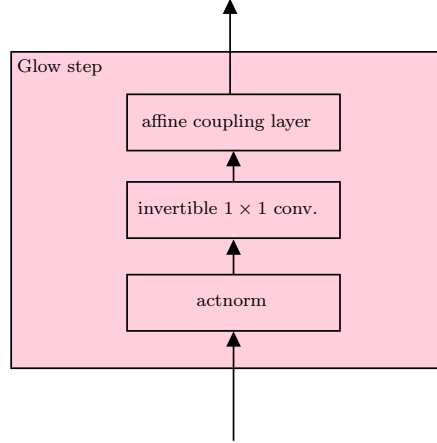


Figure 2.2: Illustration of a GLOW architecture where each transformation step follows a structured sequence: an activation normalization (Act-Norm) layer, an invertible 1×1 convolution, and an affine coupling transformation.

in vector form, then we define

$$y = Wx, \quad (2.10)$$

where W is an invertible $D \times D$ matrix. The Jacobian determinant isⁱⁱ

$$\log \left| \det \frac{\partial \mathbf{y}}{\partial \mathbf{x}} \right| = \log |\det(W)|. \quad (2.11)$$

In a third step, the affine coupling layer transforms half of the input x_2 using a learnable transformation composed of a scaling function s and a translation function t , both conditioned on the other half x_1 ⁱⁱⁱ:

$$y_2 = x_2 \cdot \exp(s(x_1)) + t(x_1), \quad (2.12)$$

while $y_1 = x_1$ is left untouched. Hence,

$$\log \left| \det \frac{\partial \mathbf{y}}{\partial \mathbf{x}} \right| = \sum_i s(x_1). \quad (2.13)$$

The invertibility of all the mentioned steps can be easily verified.

ⁱⁱFor the inverse of the matrix, we simply get $\log |\det(W^{-1})| = -\log |\det(W)|$. So taking the log-determinant in inverse direction is easy to compute.

ⁱⁱⁱAlthough looking similar to the Act-Norm, the Act-Norm is trained to normalize the data, whereas the affine coupling layer learns the complex transformation of data.

Data and Methodology

This chapter describes the procedure for extracting, preprocessing, and modeling the associated halo distributions for an H II region.

3.1 pyC2Ray-Simulation

This thesis uses the simulation data of `pyC2Ray` (Hirling et al. 2024) for training and sampling. `pyC2Ray` is a radiative transfer code designed to simulate the propagation of ionization fronts by solving the coupled equations of ionization and radiative transfer. Using it, we can create a volume of mesh-size 256, with each cell containing an ionization fraction value between $x_{\text{HI}} = 0$ and $x_{\text{HI}} = 1$. The simulation cube spans 200 Mpc/ h on each side, meaning that one pixel corresponds to approximately 0.78 Mpc/ h . Throughout this thesis, we work in comoving units but simply write Mpc instead of cMpc for readability. Alongside the ionization fraction, the simulation provides a catalog of dark matter halos, specifying the mass and three-dimensional position of each halo within the volume. While the original data are three-dimensional, we will focus on two-dimensional slices. These slices can be oriented in three fundamental planes, conventionally **xy**, **yz**, or **zx**, so that all possible spatial orientations are covered. A threshold is then applied to each slice to identify where the ionization fraction is sufficiently high to be considered part of a H II bubble. This threshold is commonly set to a value such as 0.5 so that those cells with ionization fractions exceeding 0.5 are flagged as “ionized”. The flagged cells form several connected bubbles in the slice. Each bubble can be isolated by centering a fixed-size 64×64 (assuming, no bubbles are larger than 64×64 in the redshifts of interest) patch around the bubble’s center of mass and creating a smaller two-dimensional grid. Within that patch, cells belonging to the bubble retain their ionization fraction values, while cells outside the bubble can be set to a background value, in this case zero. We enlarge the effective training set, by rotating each patch by 90, 180, or 270 degrees, or by flipping the image. This generates different views of the same patch and preserves the physics of the bubble and halo positions, while allowing a model to encounter more varied configurations.

3 Data and Methodology

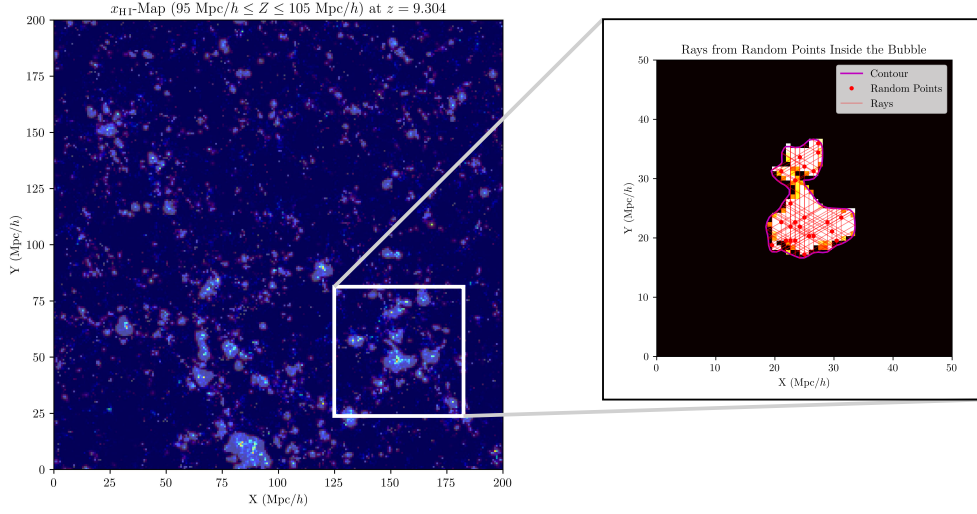


Figure 3.1: Slice of the neutral hydrogen fraction x_{HI} at redshift $z = 9.304$, covering the halo masses with z -direction $95 \text{ Mpc}/h \leq r_z \leq 105 \text{ Mpc}/h$ (left). The zoomed-in region contains one single ionized bubble, where randomly sampled points inside the bubble (red) are connected by rays to the boundary (smoothed contour). These rays illustrate the conditional vector used as the input for the model.

3.1.1 Halo Catalog Processing and Density Patches

Since each halo has a three-dimensional position (given by its x , y , z coordinates), the slicing code filters out those halos that lie sufficiently close to the slice’s plane, so that their projected positions are relevant to the corresponding two-dimensional patch. We select a small interval around the slice coordinate. In this thesis, we take:

$$r_z^i - 5 \cdot \Delta r \leq r_z \leq r_z^i + 5 \cdot \Delta r.$$

Here we examine the xy plane for a slice i in the z -direction with each slice resolution $\Delta r = 0.78 \text{ Mpc}/h$. Only halos lying in that interval are retained, and their 3D positions are then mapped onto the two-dimensional slice coordinates. We then align the halo density field with the ionization fraction slice. A complete aligned 256×256 slice is shown in Figure 3.1.

3.2 State-of-the-Art Approach

A common approach for locating ionizing sources within H II regions is to simply identify the innermost point of the bubble, under the assumption that the strongest ionizing source is unlikely to lie near its boundary. A Gaussian distribution is then employed as a simple model to account for some uncertainties.

3 Data and Methodology

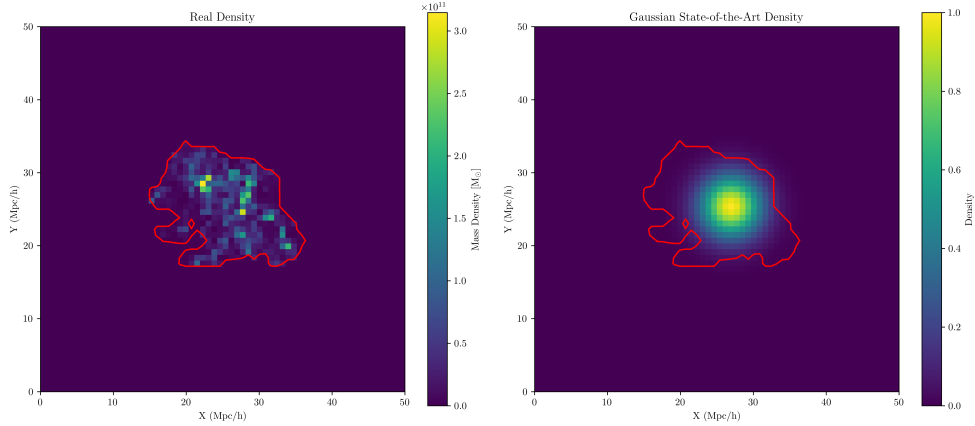


Figure 3.2: Illustration of the state-of-the-art method on a sample H II bubble. The innermost point is taken to be the ionizing source and a single Gaussian is employed to account for some uncertainties.

An illustration of this method applied to a representative H II bubble is shown in Figure 3.2. From a practical standpoint, this method is attractive for its simplicity and computational efficiency. However, as will be shown in the results (Section 4.2) and can be seen by eye from the illustration, the single-Gaussian assumption does not capture the full complexity of the ionizing source distribution, often leading to bad accuracy. In particular, the emission in real observations can exhibit strong asymmetries. More advanced techniques, such as the Normalizing Flow approach proposed in this thesis, aim to overcome these shortcomings by providing more flexible and accurate models of the ionizing source distribution.

3.3 Conditions

In order to model halo distributions in a bubble, there must be a well-defined representation of each bubble. We could use the bubbles 64×64 image as a condition. However, to reduce the total number of conditions, we first sample 24 random points inside each bubble. From each sampled point, we cast six radial rays at angles 0° , 60° , 120° , 180° , 240° , and 300° . Each ray is traced outward until it intersects the bubble boundary. The sampled point locations and the corresponding 6 ray-lengths to the contour of the bubble are then stored inside a conditional vector. This reduces the number of conditions from 64×64 to 24×8 (24 sampled points, for each point we store the position of the point (x, y) and the 6 ray lengths). Combined with the total halo mass within the bubble (which will be discussed later in the thesis), this forms the conditional vector that serves as the input to our model.

3.4 Geometry Measurement & Metrics

In this section, we introduce the key shape metrics that characterize each extracted bubble in our dataset. The following shape measurements are relevant for this thesis (Belongie et al. 2002; van der Walt et al. 2014; Wadell 1935; Zhang & Lu 2004):

Compactness

Compactness is defined as the ratio of the area of an object to the area of a circle with the same perimeter:

$$\text{compactness} = \frac{4\pi (\text{area})}{(\text{perimeter})^2}. \quad (3.1)$$

The metric takes a maximum value of 1 for a circle and $\frac{\pi}{4}$ for a square.

Elongation

The elongation is computed as

$$\text{elongation} = \frac{\text{bounding box width}}{\text{bounding box length}}. \quad (3.2)$$

Values close to 1 indicate isotropic-like shapes (roughly circular or square), while significantly higher or lower values (e.g., 0.3 or 2.5) reflect elongated structures. Since width and length depend on the orientation of the bubble, elongation values of x and $1/x$ should be considered physically equivalent.

Solidity

The ratio of a region’s area to the area of its convex hull:

$$\text{solidity} = \frac{\text{area}}{\text{convex Area}}. \quad (3.3)$$

A value of 1 signifies a solid object and a value less than 1 signifies an object having an irregular boundary or containing holes. Objects with solidity around 0.7–0.8 typically can be starfish-like shapes or crescents.

Volume

Under the assumption of uniform slice thickness, we use the approximate volume

$$\text{volume} \approx (\text{area in 2D}) \times (\text{slice thickness}), \quad (3.4)$$

throughout this thesis.

Sphericity

Sphericity measures the degree to which an object approaches the shape of a sphere:

$$\text{sphericity} = \frac{R_{\text{inscribed}}}{R_{\text{circumscribed}}}, \quad (3.5)$$

where $R_{\text{inscribed}}$ is the radius of the inscribed circle in the object, and $R_{\text{circumscribed}}$ is the radius of the circle circumscribing the object. Sphericity takes a maximum value of 1 and a minimum value of 0. For a circle, the value is 1. Sphericity is $= \frac{\sqrt{2}}{2} \approx 0.71$ for a square and $= \frac{1}{2} = 0.50$ for an equilateral triangle.

Eccentricity

Eccentricity is the ratio of the length of the short (minor) axis to the length of the long (major) axis of an object:

$$\text{eccentricity} = \frac{\text{length of minor axis}}{\text{length of major axis}}, \quad (3.6)$$

given as a value between 0 and 1 (circle).

Figure 3.3 shows the distribution of these shape characteristics across our entire bubble dataset.

In addition to these shape descriptors, we evaluate the performance of our model using the R^2 score (coefficient of determination). The R^2 score provides a measure of how well the model's predictions match the observed data, defined as (Heinisch 1962)

$$R^2 = 1 - \frac{\sum_i (y_{\text{true},i} - y_{\text{pred},i})^2}{\sum_i (y_{\text{true},i} - \bar{y}_{\text{true}})^2}, \quad (3.7)$$

where $\bar{y}_{\text{true}}$ is the mean of y_{true} . We employ a row/column-based method of computing R^2 for two-dimensional data (matrices), where we compare row sums and column sums separately and then take the average. This approach allows us to capture the overall agreement of predicted and true distributions along both dimensions. For the performance of the **SegU-Net** (see Section 4.4), we will furthermore use the Matthews Correlation Coefficient (MCC) which is a metric often used in binary classification tasks (Matthews 1975):

$$\text{MCC} = \frac{\text{TP} \times \text{TN} - \text{FP} \times \text{FN}}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}}, \quad (3.8)$$

where TP, TN, FP, and FN represent the counts of true positives, true negatives, false positives, and false negatives, respectively. A MCC coefficient of 1 indicates a perfect prediction and -1 indicates total disagreement.

3 Data and Methodology

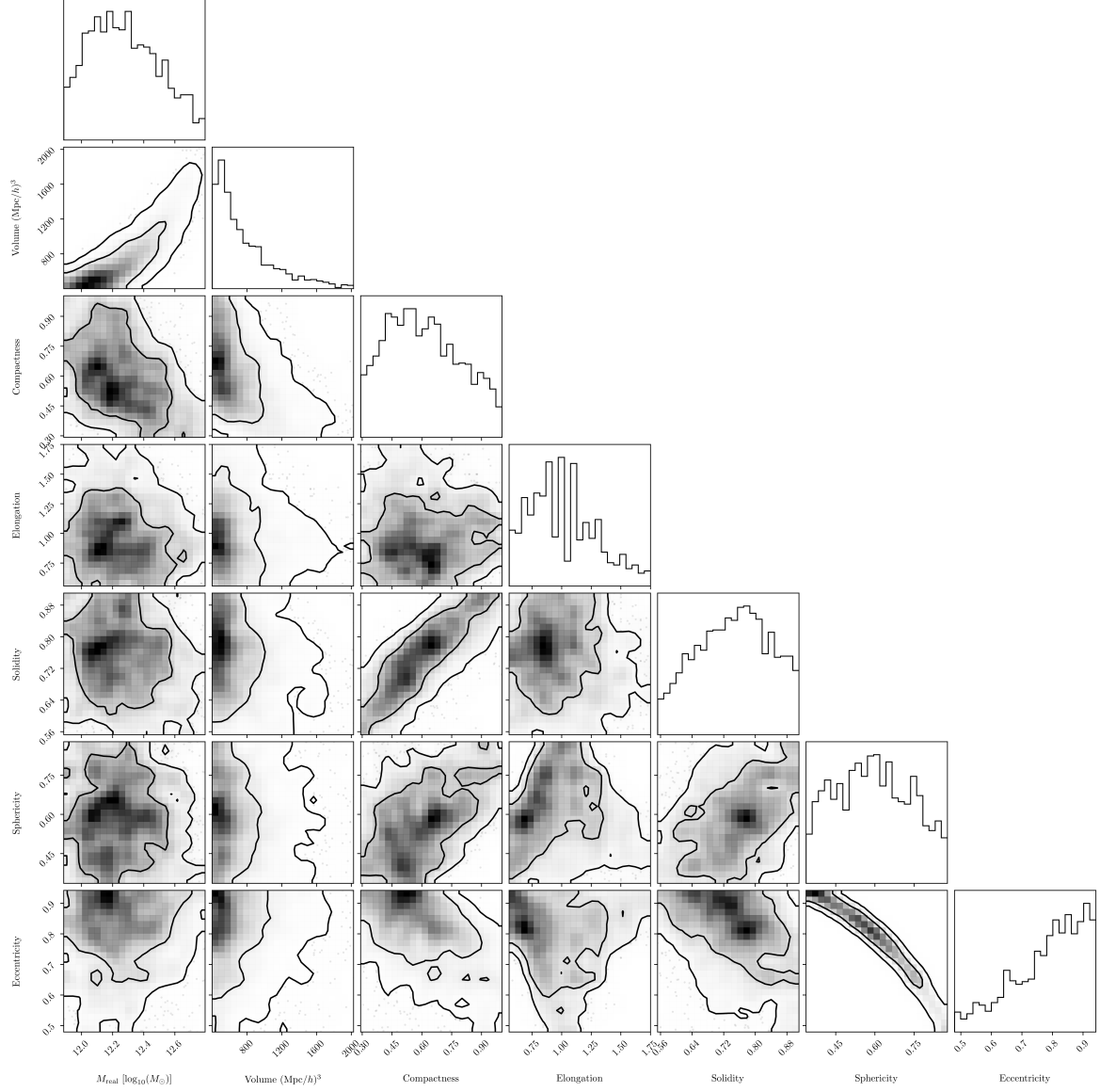


Figure 3.3: Distribution of shape metrics across an extracted bubble dataset. The figure shows the relationships between compactness, elongation, solidity, volume, sphericity, and eccentricity, highlighting the geometric diversity of the bubbles.

Results and Discussion

4.1 Training

For the Normalizing Flow model, we use the GLOW-architecture, which was described in chapter 2. We take 60 density patches and their transformations introduced in subsection 3.1.1, along with their associated conditional vectors, as training batches. The core objective is to maximize the likelihood of the observed data under the flow. Equivalently, we minimize the negative log-likelihood (NLL). For a batch of size N , each containing pairs $(\mathbf{x}_i, \mathbf{c}_i)$ of coordinates and condition vectors, the loss function is given by

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \log p(\mathbf{x}_i | \mathbf{c}_i), \quad (4.1)$$

where $p(\mathbf{x}_i | \mathbf{c}_i)$ is computed through the GLOW transformations that map $\mathbf{x}_i$ to the latent space (see Equation (2.7)). For the base distribution, we choose a simple two-dimensional Gaussian. We train the model using AdamW optimizer, suggested by Loshchilov & Hutter (2019). A step-based learning-rate scheduler decreases the learning rate every 100 epochs. A randomly chosen 20% of the data is reserved for validation.

Figure 4.1 shows both the training and validation losses over the course of $N_{\text{epochs}} = 3170$ epochs. We see that the model converges steadily, with no major signs of overfitting; the validation loss closely tracks the training loss and eventually stabilizes. Starting around epoch 1500, we observe that the validation loss no longer decreases significantly and begins to diverge slightly from the training loss. To avoid overfitting, we terminate the training after 3000 epochs. We summarize the key hyperparameters of our GLOW-based flow in Table 4.1. For reproducibility, the GitHub repository (Krizic 2025) provides the training implementation, mock data, and sampling scripts.

4 Results and Discussion

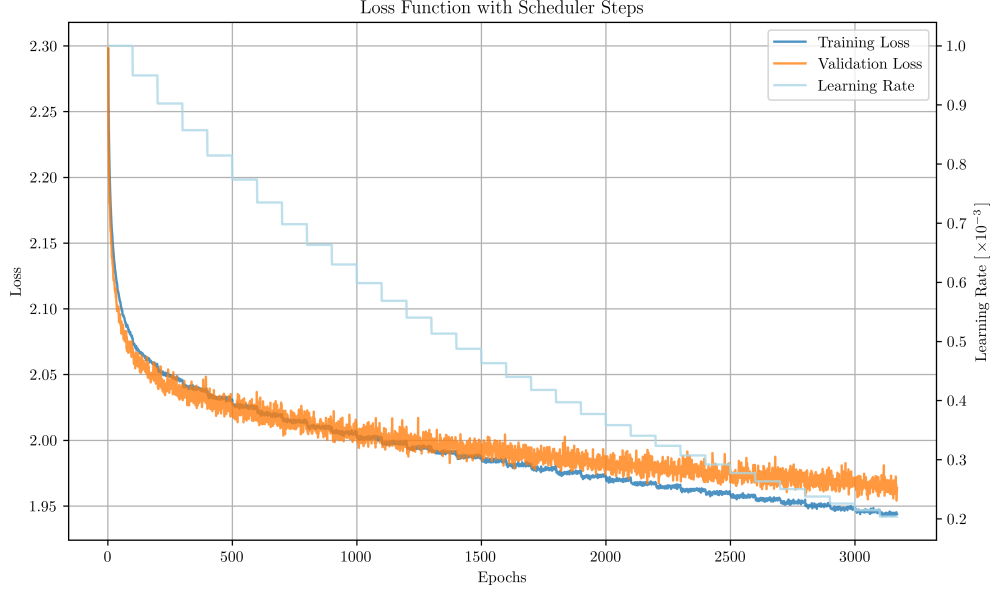


Figure 4.1: Training (blue) and validation (orange) losses plotted against the number of epochs. Note that around epoch 1500, the validation loss diverges slightly from the training loss.

Parameter	Value	Notes
N_{epochs}	3170	—
Number of GLOW steps	6	Each step consists of: • ActNorm • 1x1 Convolution • Affine Coupling
Number of Nodes	256	Used in Affine Coupling layers
Batch size	256	—
Initial learning rate	10^{-3}	Decays by a factor of 0.95 every 100 epochs
Optimizer	AdamW	Loshchilov & Hutter (2019)
Number of Simulated Points in NF	1000	—
Number of Points per Bubble	24	See Section 3.3
Number of Rays per Point	6	—

Table 4.1: Overview of the GLOW-based flow hyperparameters used in training.

4 Results and Discussion

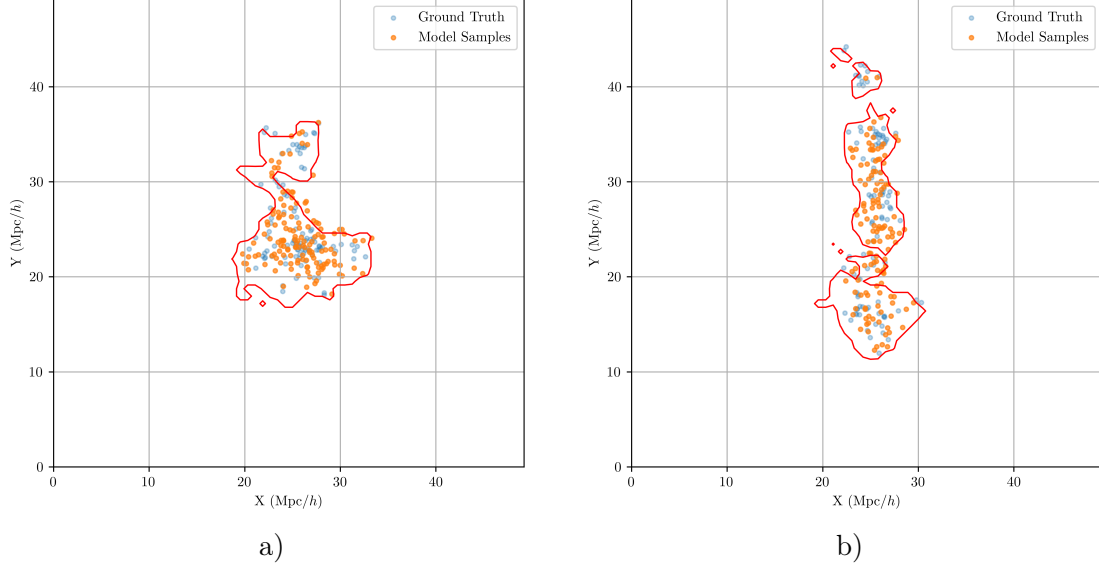


Figure 4.2: Example comparison of ground truth vs. modeled halo densities for two different bubbles. The blue samples show the real, ground truth, halo distributions by randomly selecting points with probability proportional to halo mass densities at these locations, while the orange dots show the corresponding modeled samples produced by the GLOW network. In red, we illustrate the contours of the corresponding bubbles.

4.2 Sampling of Halo Distributions

Assuming we already know the mass of the haloes inside the bubble, one of the most direct ways to evaluate our model is to compare real halo maps (from the C^2Ray halo catalog) against those generated by the flow. From the halo catalog, we randomly create 100 points, where the probability of creating a point at a position is given by the accumulated mass of the halos at this point, and sample 100 points from our model. One sampling result is shown in Figure 4.2. In these examples, the generated distributions show clustering patterns and regions similar to those in the real data. This is illustrated more clearly in Figure 4.3, where the ground truth halo density and the modeled halo density for the bubble from Figure 4.2 are shown as density fields. High-density spots appear at locations consistent with the ground truth density. Although minor discrepancies can occur (e.g. in smaller substructures), the overall visual agreement suggests that the model captures the main halo density trends.

While visual comparisons show that our model replicates key density features of the halo distribution, a more quantitative assessment can be made by examining how well the modeled halo density fields match the true data in terms of the R^2 -score (see Equation 3.7). In particular, we assess how R^2 varies with different bubble and halo shape descriptors introduced in Section 3.4 and also compare across different redshifts. Fig-

4 Results and Discussion

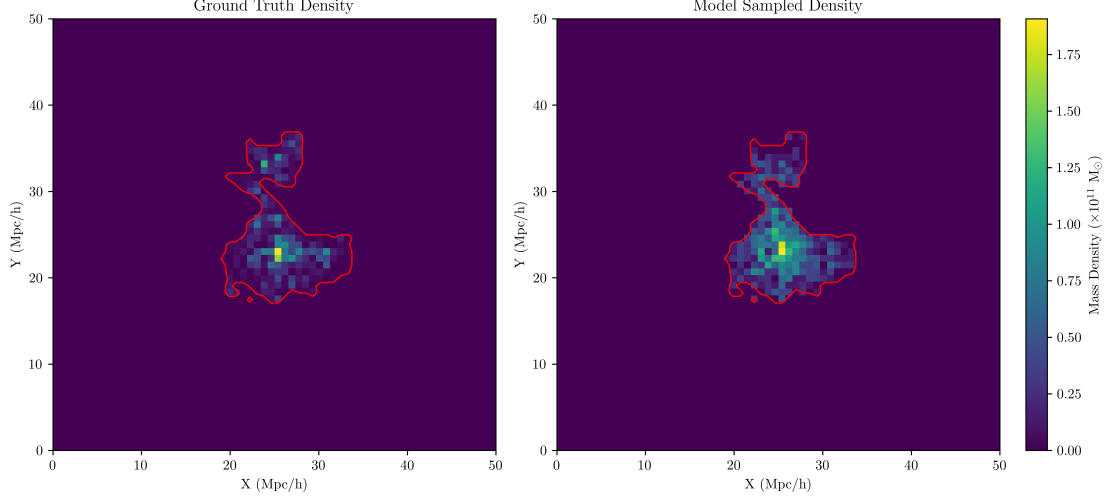


Figure 4.3: Side-by-side comparison of the 2D halo density fields for ground truth (left) and GLOW-generated (right) samples. The color scale indicates the halo mass density in a pixel, and the red contour marks the boundary of the corresponding bubble.

ure 4.4 shows the results for redshift $z \sim 9.304$.

Notably, the model maintains strong performance (with an average R^2 -Value of $\langle R^2 \rangle \approx 86.11\%$) across a broad spectrum of shapes (especially sphericity, solidity, compactness and eccentricity), suggesting that differences in shape have only a modest effect on the predictive accuracy. In terms of size, most of our dataset features relatively small volumes. However, the model still achieves an average of $\langle R^2 \rangle \approx 80.45\%$ if we only take bubbles into account with volumes exceeding 2000 (Mpc/h)^3 . Figure 4.5 shows the R^2 -value distribution at $z \sim 9.304$ for different volume bins.

Finally, we still have to benchmark our results against the state-of-the-art Gaussian-based halo modeling approach (introduced in Section 3.2). In Figure 4.6, we plot R^2 vs. bubble volume for the Gaussian approach. We obtain an average R^2 value of $\langle R^2 \rangle \approx 43.62\%$. By comparing this value to the $\langle R^2 \rangle \approx 86.11\%$ obtained using our flow-based model, we see that the GLOW model outperforms the Gaussian state-of-the-art approach.

In summary, the R^2 -value analysis shows the model’s ability to handle not only simple or spherical bubbles but also those with more eccentric shapes significantly better than the simple state-of-the-art approaches.

4 Results and Discussion

R^2 vs. Shape Metrics, $\langle R^2 \rangle \approx 86.11\%$

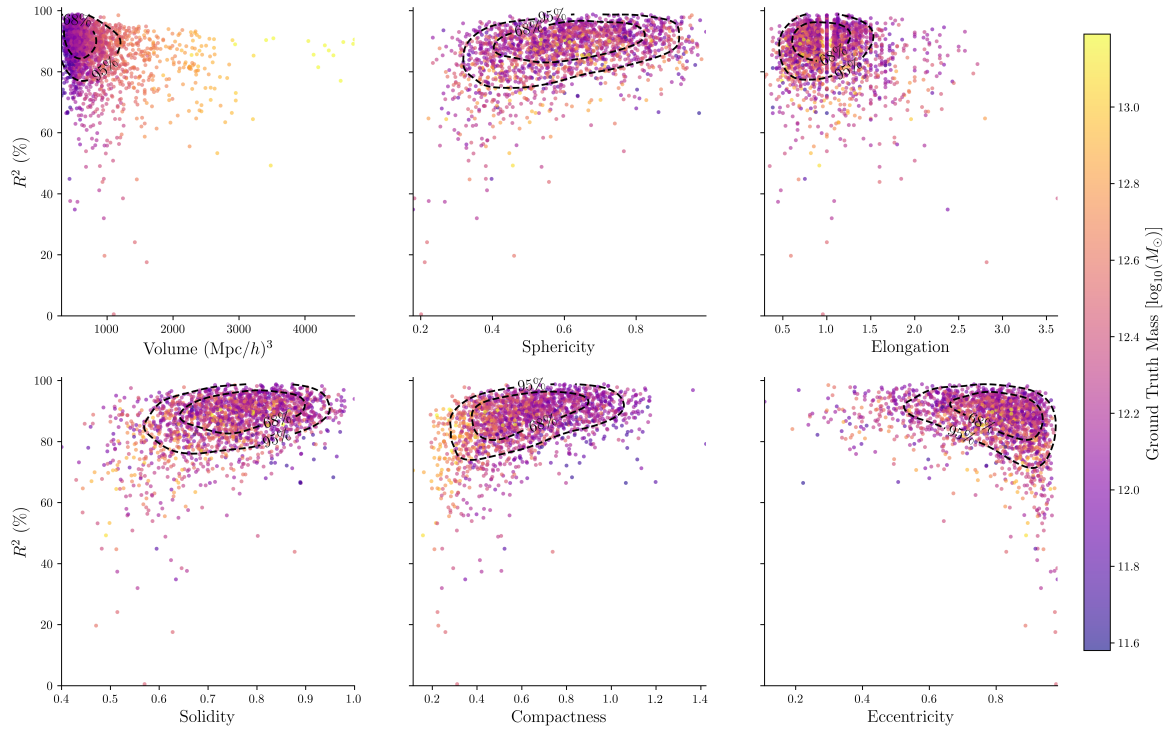


Figure 4.4: R^2 values plotted against several different shape metrics at $z \sim 9.304$. Each point represents an individual bubble.

4 Results and Discussion

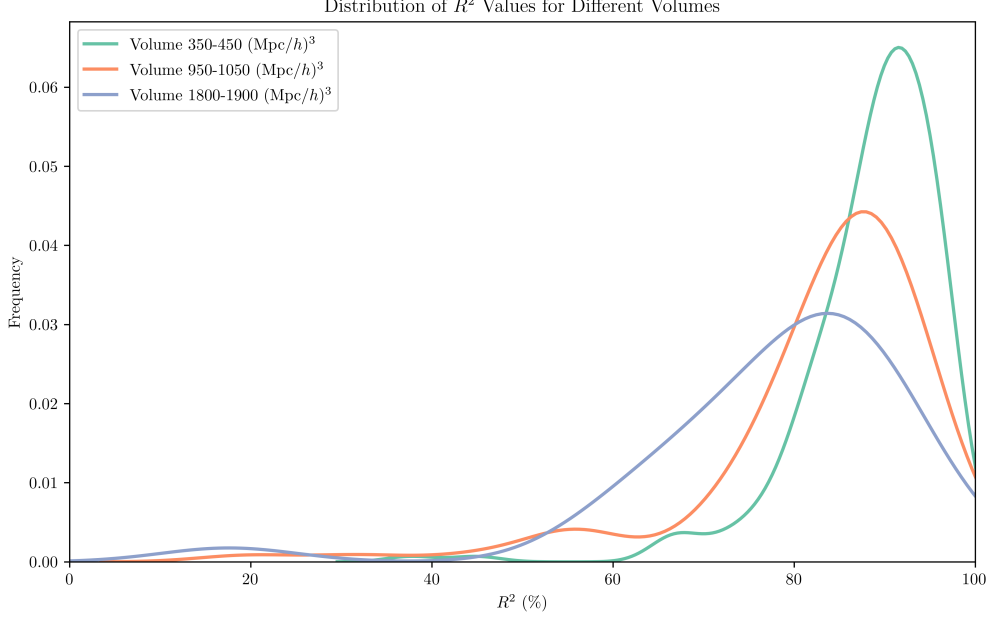


Figure 4.5: Distributions of R^2 values for different volume bins. Bubbles are grouped by their approximate volume into three different size categories to illustrate, how bubble size affects the model’s accuracy.

4.3 Total Halo Mass Prediction

Until now, we viewed the mass as a condition of the bubble, although we actually do not know the total mass from the contour of the bubble. However, we can also infer the total mass for a given bubble by scanning through possible values and identifying which yields the highest log-likelihood. Concretely, we build a set of candidate masses over an interval around the “real” mass (for example, spanning one or two orders of magnitude above and below), and for each candidate mass, we first construct a density map by sampling from the flow, using the candidate mass in the conditional vector. We then compare the resulting density map to the actual halo density in the bubble through the log-likelihood. By repeating this procedure for all candidate masses, we can select the mass that achieves the maximum log-likelihood as the total mass. Figure 4.7 illustrates one such mass scan, indicating how the log-likelihood varies across the candidate range. Typically, the curve shows a peak near the true mass, confirming that the flow has learned how total mass influences the distribution of halos in the bubble.

Overall, this method allows the mass parameter to be treated as an unknown and inferred solely from the halo arrangement. Thereby we demonstrated, that our conditional GLOW can serve not only as a generative tool but also as a mass estimator in HII-region analysis. Figure 4.8 shows the correlation between the real and modeled halo masses, demonstrating this mass scan procedure.

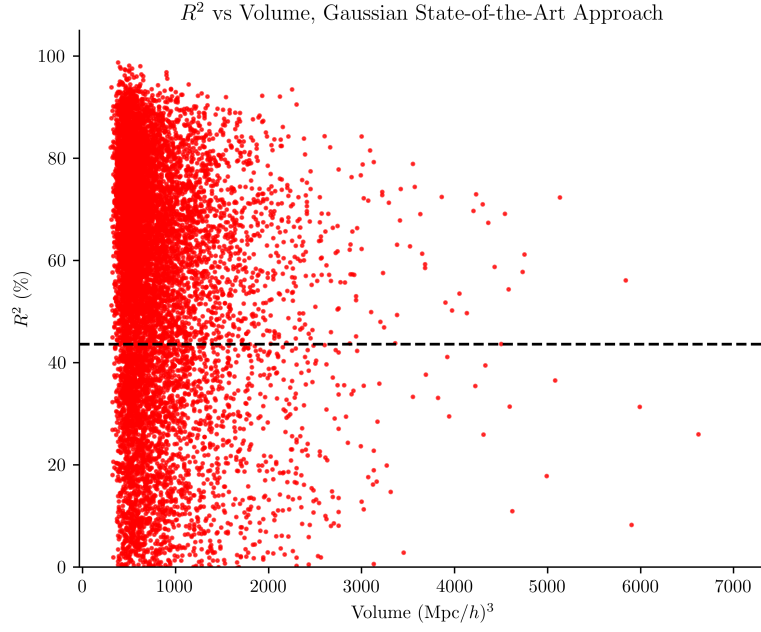


Figure 4.6: R^2 values of halo predictions from the Gaussian-based state-of-the-art approach, plotted against bubble volume (at $z \sim 9.304$). We get an average of $\langle R^2 \rangle \approx 43.62\%$ (black dashed line). This value serves as a benchmark for our GLOW model.

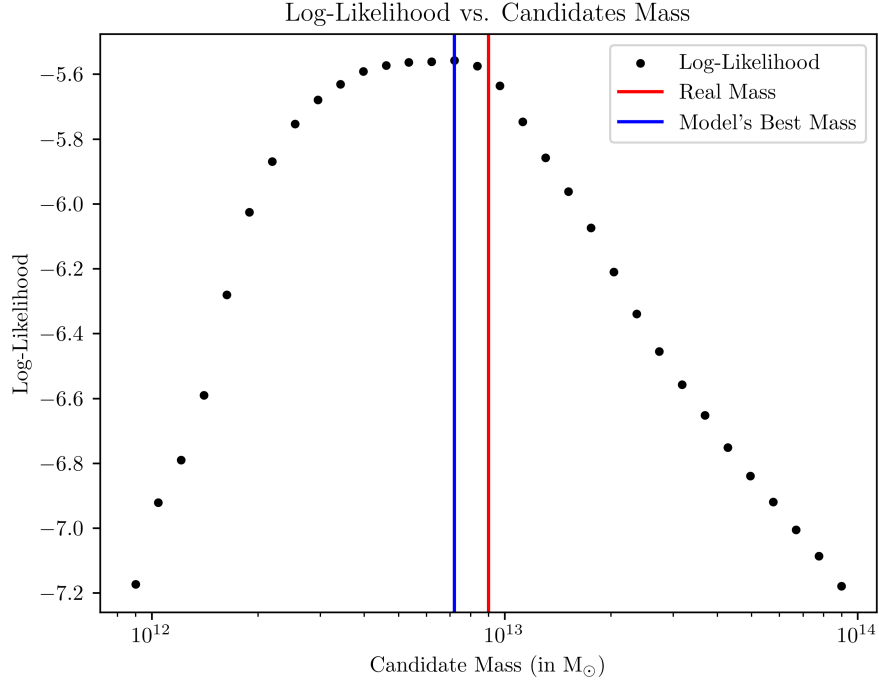


Figure 4.7: Example of scanning over candidate masses for the same bubble as in Figure 4.3. The plot shows the log-likelihood obtained by our flow model versus the candidate mass. The red vertical line indicates the real mass measured directly from the simulation, while the blue line marks the model’s best estimate, respectively the mass maximizing log-likelihood.

4 Results and Discussion

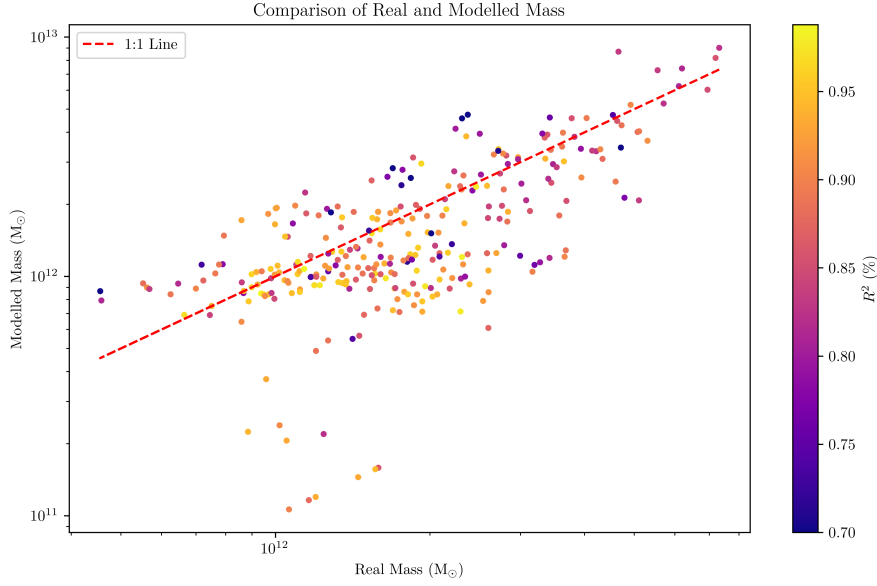


Figure 4.8: Correlation between real and modeled halo masses with the R^2 score as colorbar. The modeled mass was estimated using the maximum log-likelihood method by evaluating the log-likelihood of density distributions conditioned on 100 different candidate masses.

Figure 4.9 presents the relationship between the ionizing photon production rate, $\dot{N}_\gamma$ (see Section 1.2.1), and the bubble volume. The model’s prediction (black data points) agrees with the ground truth data, and the linear mean fits of the model and ground truth data (green and orange dashed lines) confirm that both are well-matched overall. The vertical dashed blue line in the plot marks the telescope resolution threshold. The observed scatter on the left side of this line is therefore irrelevant for the SKA-Low, since such bubbles cannot be well resolved. However, despite this scattering at lower volumes, the model effectively reproduces the ionizing production rate.

4.4 Performance on Noisy Data

To evaluate the robustness of our model in realistic scenarios, we test its performance on noisy data. In order to simulate the noise, we use the `tools21cm` package (Giri et al. 2020), which provides realistic noise realizations corresponding to the SKA-Low telescope. First, we apply noise to a clean `C2Ray` snapshot, generating 100 differently-noised versions of the same slice through random re-sampling. Then we extract the ionized bubble from each noisy map using the `SegU-Net` network (Bianco et al. 2021), a segmentation model based on a U-Net architecture. `SegU-Net` has been pre-trained on `py21cmFAST` data and demonstrates strong performance in locating ionized regions within noisy 21 cm maps. Since our GLOW model is trained on `C2Ray` data, we focus on a single, clearly isolated

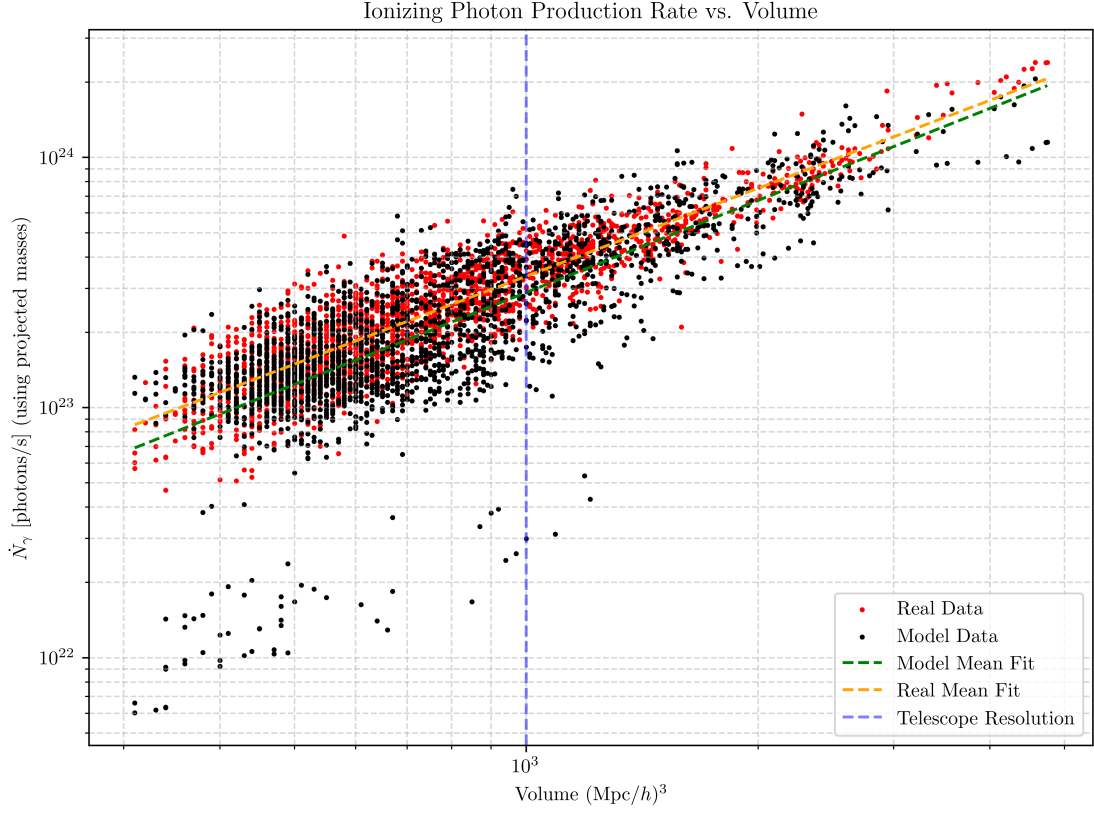


Figure 4.9: Ionizing photon production rate $\dot{N}_\gamma$, plotted against bubble volume for both ground truth (“real”) data (red) and model predictions (black). The orange and green dashed lines show the linear fits of the ground truths (orange) and the model samples (green), while the vertical dashed blue line marks the SKA-Low-telescope resolution threshold. Despite scatter at lower volumes, the model reproduces the general relation observed in the ground truth data.

bubble, where **SegU-Net** performs well (with a $\text{MCC} \approx 80\%$). This bubble is extracted under all the 100 distinct noise conditions. For each of the 100 realizations, we compute the R^2 value between the predicted and the ground truth density and additionally compute the key shape descriptors again. The distribution of all metrics is summarized in a corner plot in Figure 4.10. Overall, the **SegU-Net** + GLOW model achieves a performance ($\langle R^2 \rangle \approx 94.23\%$) that is comparable to applying the GLOW model directly to the ground truth data ($\langle R^2 \rangle \approx 96.04\%$).

4.5 Performance on Different Astrophysics and Redshifts

In addition to the case at $z \sim 9.304$ (as until now), we evaluate our GLOW model at various redshifts and astrophysical parameters to show its performance under different bubble growth conditions. Table 4.2 shows the average R^2 -scores at three different redshifts for the same astrophysical scenario (**S1SA**; the models are explained in more detail in Footnote i in Section 1.2.1).

Redshift	$z \sim 8.895$	$z \sim 9.304$	$z \sim 9.922$
$\langle R^2 \rangle$	85.16%	86.11%	86.23%

Table 4.2: Average R^2 -scores of the GLOW model for scenario **S1SA** at different redshifts.

These results reinforce the trend that the model tends to perform slightly better at higher redshifts, where the bubbles are smaller and more compact, thus making it easier for the Normalizing Flow to capture the halo distributions. As redshift decreases toward $z \sim 8$, bubble shapes become larger and often more irregular, which can lower the R^2 -score by a few percentage points. Nevertheless, even for these more challenging low-redshift bubbles, the model still achieves R^2 -values over 80%.

Table 4.3 compares the performance of GLOW across several different reionization models at redshift $z \sim 9.304$. Despite different source and sink models, the GLOW model achieves consistently high predictive accuracy.

Reionization Model	S1SA	S1SB	S1SC	S2SA	S3SA
$\langle R^2 \rangle$	86.11%	88.16%	88.24%	85.10%	82.09%

Table 4.3: Average R^2 -scores of the GLOW model for different astrophysical parameters at redshift $z \sim 9.304$.

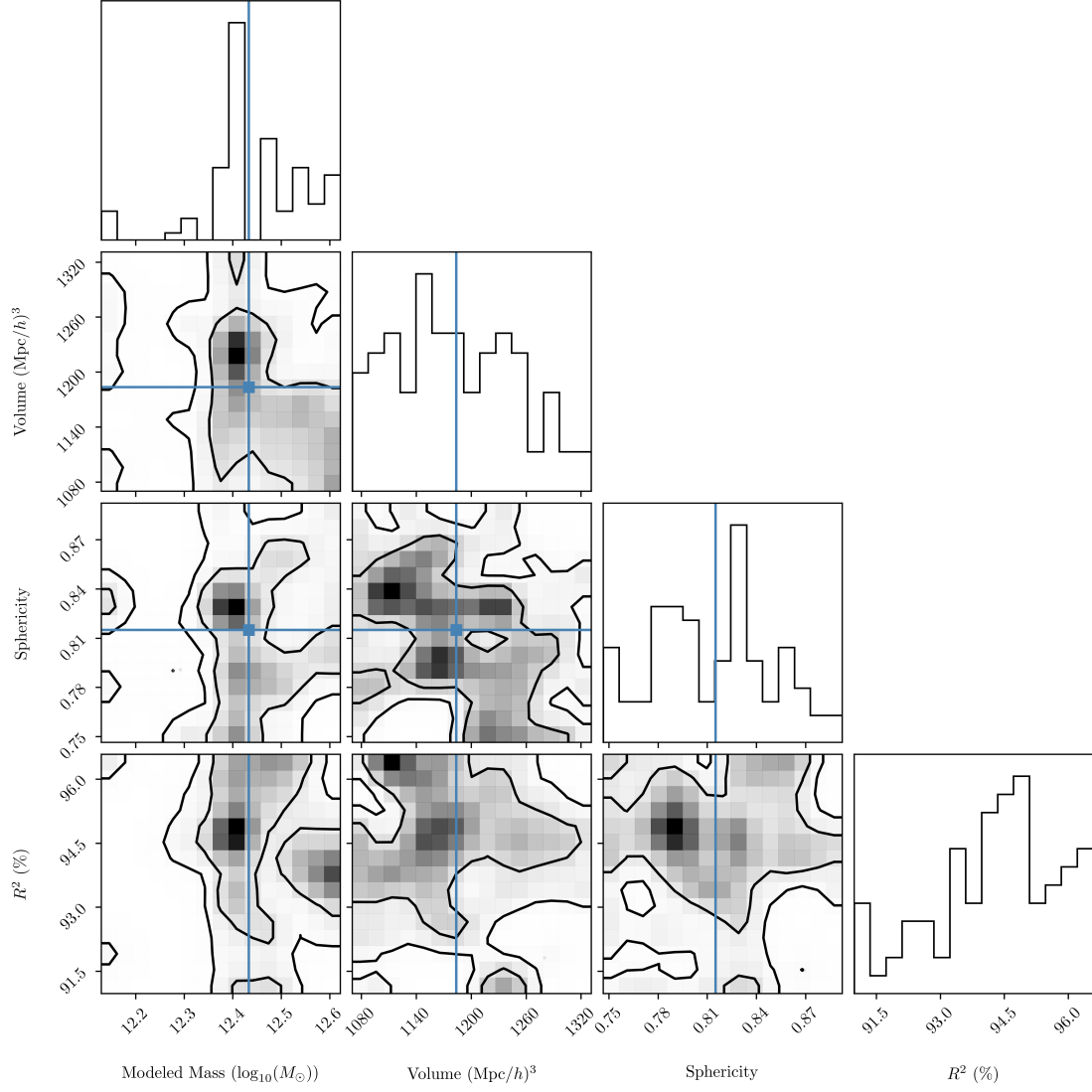


Figure 4.10: Corner plot showing the distribution of R^2 values and shape descriptors for a single bubble subjected to 100 different noise realizations using **SegU-Net** and the GLOW model. The blue lines show the ground truth (i.e. only applying the GLOW model to the ground truth data).

Conclusion

This thesis presents a novel machine learning approach to modeling the distribution of ionizing sources within H II regions during the EoR, using conditional Normalizing Flows based on the GLOW architecture. By conditioning on a set of 24 interior points with six radial distances to the bubble boundary (see Section 3.3), along with the total halo mass, the model learned to reproduce complex and often asymmetric halo distributions inside H II regions. Unlike simpler state-of-the-art models, such as placing a Gaussian around the innermost bubble point introduced in Section 3.2, the flow-based method achieves significantly higher accuracy, with average R^2 scores exceeding 80% across a wide range of bubble shapes, sizes, and astrophysical conditions (see Section 4.2 and Section 4.5), whereas the state-of-the-art approach achieves an average of $R^2 \lesssim 44\%$.

Furthermore, by combining the model with segmentation networks such as SegU-Net, we show in Section 4.4 that the method generalizes well to noisy data resembling SKA-Low observations, achieving near-identical performance to that obtained from clean simulation slices.

In addition to its generative capabilities, we explained in Section 4.3 how the model can infer the total halo mass of a given bubble by scanning over candidate values and selecting the one that maximizes the log-likelihood. This capability makes the method suitable not only for reconstructing source distributions but also for estimating physically meaningful quantities from observational data.

While this work focused on two-dimensional slices, extending the model to three dimensions is a natural and promising next step. Such an extension could further improve accuracy and better capture line-of-sight effects, which may become relevant for future 3D halo density reconstructions. Additionally, adding different shape characteristics to the conditional vector may offer improved performance on larger or more irregular bubbles. Finally, while the current model was trained on simulations with specific astrophysical assumptions, the approach is flexible and could be retrained or fine-tuned for other reionization models or cosmological simulations.

Ultimately, the motivation behind this work is to support upcoming cross-observational studies that combine 21 cm data from SKA-Low with infrared galaxy observations from JWST. By providing a data-driven way to connect the geometry of ionized regions to the underlying ionizing sources, the model can help identify where to point JWST within SKA-detected H II bubbles, thus enabling a more efficient and targeted search for high-redshift galaxies.

5 Conclusion

In summary, this thesis demonstrates that conditional Normalizing Flows offer a powerful and promising tool for modeling ionizing source distributions during cosmic reionization. By combining Normalizing Flows and astrophysical simulations, the method opens an entirely new avenue for analyzing incoming 21 cm observations and for connecting the geometry of H II regions to the galaxies that shaped them.

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